Milestone 2 Automated Feature Extraction and Matching

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https://github.com/Wayne0758/Computer-Vision

SuperPoints:

Automated Feature Extraction:

1. Points selection of IMG_1223(A):

2. Points selection of IMG_1224(B):

3. Points selection of IMG_1226(C):

Homography Estimation and Validation:

```
Estimated Homography Matrix 1:
 [[ 1.83656849e+00 -3.41910060e-01 -8.71889628e+02]
 [ 4.00212265e-01  1.18145426e+00 -7.26930507e+02]
 [ 1.65675063e-04 -9.09487529e-05  1.00000000e+00]]
Estimated Homography Matrix 2:
 [[ 1.46995597e+00  3.67075199e-01  4.34533996e+02]
 [ 2.83806754e-01  1.49534503e+00 -6.68532084e+02]
 [ 1.71892276e-04  1.52926215e-04  1.00000000e+00]]
Estimated Homography Matrix 3:
 [[ 7.20075792e+00 -1.41875514e-01 -4.27477054e+03]
 [ 2.53689542e+00  4.09645520e+00 -5.43863068e+03]
 [ 1.56011358e-03  1.59028828e-05  1.00000000e+00]]
Validation A-B: Point in A: [1609. 1543.] Point in B: [1399. 1586.] Reconstructed Point: [1381.220392098849, 1544.9630584356992]
Validation B-C: Point in B: [590.5 646. ] Point in C: [1198.    386.25] Reconstructed Point: [1282.7484025826654, 387.4460662747463]
Validation A-C: Point in A: [2994.    494.75] Point in C: [3098.  744.] Reconstructed Point: [3031.2672142518263, 736.6908578505097]
```

Fundamental Matrix and Epipolar Geometry:

1. F matrices

```
Estimated Fundamental Matrix F1:
 [[-6.08865386e-09  2.43165119e-07 -5.06218438e-04]
 [-1.11374095e-09  7.94724970e-09  1.59049058e-03]
 [ 5.80031641e-05 -2.23352737e-03  1.00000000e+00]]
Estimated Fundamental Matrix F2:
 [[-3.04291895e-09  1.67311919e-07 -3.18427096e-04]
 [ 6.31999465e-08  4.67178113e-08  1.34795563e-03]
 [-7.75015565e-05 -1.83163634e-03  1.00000000e+00]]
Estimated Fundamental Matrix F3:
 [[-5.84739252e-09  1.05866529e-07 -2.07244260e-04]
 [ 1.39813490e-07  1.03342696e-08  4.67462798e-04]
 [-2.18772507e-04 -9.96768575e-04  1.00000000e+00]]
```

2. Visualization of epipolar lines Before RANSAC

3. Visualization of epipolar lines After RANSAC

Observation:

- The mapped points generally aligned well on planar surfaces.
- Small deviations occurred around non-planar structures, which is expected due to the homography assumption.
- Most points aligned closely to their corresponding epipolar lines, validating the estimated F matrices.
- Some mismatches appeared due to noise and repetitive patterns.

DISK:

Automated Feature Extraction:

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3. Points selection of IMG_1226(C):

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```
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```

2. Visualization of epipolar lines Before RANSAC

3. Visualization of epipolar lines After RANSAC

Observation:

- DISK showed denser keypoints in textured regions.
- Mapped points showed good local accuracy, but occasionally missed larger structural alignment
- DISK produced more matches overall, but slightly noisier.
- Geometric verification via RANSAC helped prune mismatches effectively.

SIFT:

Automated Feature Extraction:

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```

2. Visualization of epipolar lines Before RANSAC

3. Visualization of epipolar lines After RANSAC

Observation:

- SIFT detected stable keypoints especially in textured and corner regions.
- Homography mapping was very reliable for planar areas.
- Very strong alignment between points and epipolar lines.
- Few outliers remained after RANSAC pruning.

Methods Comparison:

- Disk: Faster and more lightweight, but produced noisier matches.
- SIFT: More computationally expensive but very reliable for textured scenes.
- SuperPoint: Balanced speed and robustness well, good for general purposes.

Conclusion:

In this milestone, we automated feature extraction and matching using SuperPoint, Disk, and SIFT. Homography estimation and epipolar geometry validation showed that SuperPoint and SIFT provided the most reliable matches. Disk detected more features but introduced more noise. After estimating the fundamental matrix and visualizing the epipolar geometry, we observed that the majority of the keypoints align closely with their corresponding epipolar lines. This indicates that the estimated fundamental matrix is geometrically consistent with the true camera motion and scene structure. The alignment between points and lines confirms the effectiveness of the matching and geometric verification steps, with minimal deviation from the predicted epipolar constraints. Overall, the results demonstrate accurate and reliable two-view geometry. RANSAC geometric verification effectively removed incorrect matches. These automated pipelines pave the way for scalable 3D reconstruction and structure-from-motion tasks.